

Mutual Fund Analytics Capstone Project

Bluestock Fintech Capstone Program

Prepared By

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Executive Summary

The Mutual Fund Analytics Capstone Project was developed to analyze mutual fund performance, investor behavior, benchmark tracking, and risk metrics using real-world financial datasets. The project integrates data engineering, data analytics, and business intelligence techniques to generate meaningful investment insights.

The solution includes an ETL pipeline, SQLite database design, exploratory data analysis, performance analytics, investor analytics, benchmark comparison, advanced risk analysis, and an interactive Power BI dashboard. The project demonstrates how data-driven techniques can support investment decision-making and portfolio evaluation.

Problem Statement

Mutual fund companies and investors generate large volumes of transactional and market-related data. However, raw data alone does not provide meaningful insights for investment decisions.

The objective of this project is to build a complete analytics solution that transforms raw mutual fund datasets into actionable insights through data processing, performance measurement, risk analysis, and dashboard visualization.

Project Objectives

The primary objectives of this project are:

- Build an end-to-end ETL pipeline.
- Create a centralized SQLite database.
- Perform exploratory data analysis on mutual fund datasets.

- Calculate key performance metrics such as Sharpe Ratio, Sortino Ratio, Alpha, and Beta.
 - Analyze investor demographics and investment behavior.
 - Compare mutual funds against benchmark indices.
 - Develop advanced risk analytics including VaR and CVaR.
 - Create an interactive Power BI dashboard for business users.
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Data Sources

The project utilizes multiple datasets related to mutual funds and investor transactions.

Datasets used include:

- Net Asset Value (NAV) History
- Mutual Fund Master Data
- Investor Transaction Data
- Benchmark Index Data
- SIP and AUM Information
- Portfolio Holdings Data

These datasets provide comprehensive information regarding fund performance, risk, investor behavior, and benchmark tracking.

ETL Pipeline

The ETL (Extract, Transform, Load) process was implemented to prepare data for analysis.

Extract

Raw CSV datasets were collected from multiple sources and loaded into Python.

Transform

Data cleaning activities included:

- Handling missing values
- Standardizing column names
- Converting date formats

- Removing duplicates
- Validating data consistency

Load

The transformed data was loaded into a SQLite database for efficient querying and analytics.

The ETL pipeline improved data quality and enabled downstream analytical processes.

Database Design

SQLite was used as the primary storage layer for this project.

Major database tables include:

- fact_nav
- fact_transaction
- benchmark_data
- fund_master

Relationships were established using fund identifiers and date fields.

The database design ensured efficient storage, retrieval, and analysis of mutual fund data.

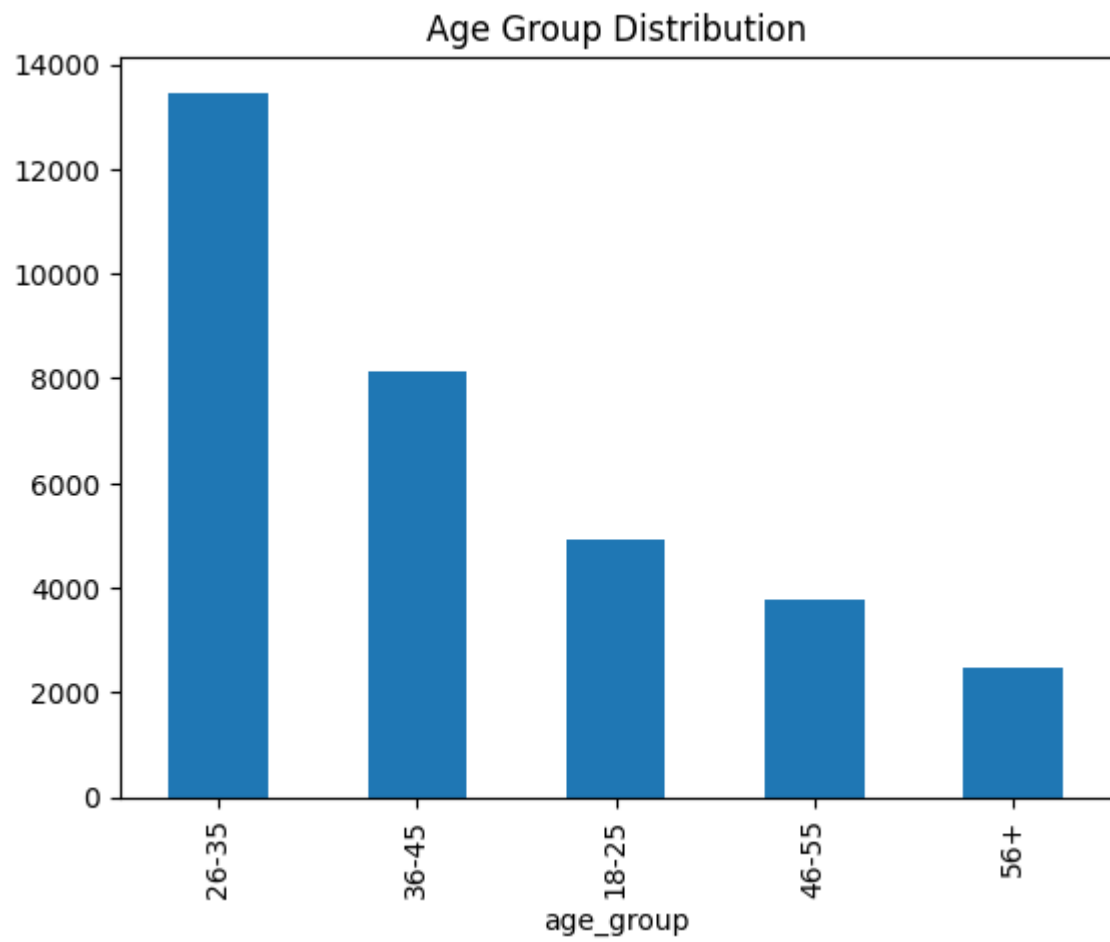
Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand dataset characteristics and identify important trends.

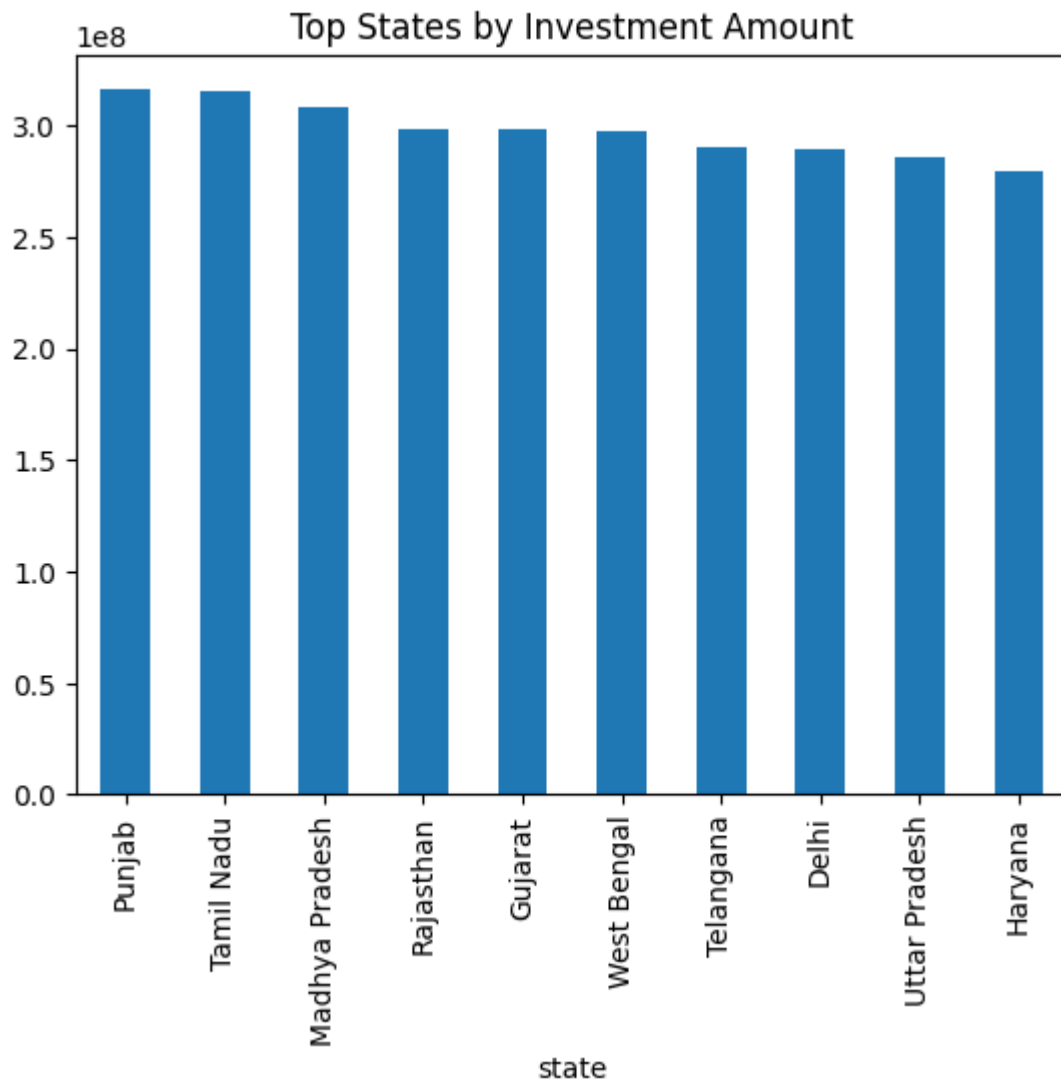
The following analyses were conducted:

- Age Group Distribution
- Gender Distribution
- City Tier Distribution
- Transaction Type Distribution
- Income Distribution
- State-wise Investment Analysis

The EDA process revealed important demographic and geographic investment patterns.



Age Group Distribution



Top States by Investment Amount

Performance Analytics

Performance analytics were conducted to evaluate mutual fund returns and risk-adjusted performance.

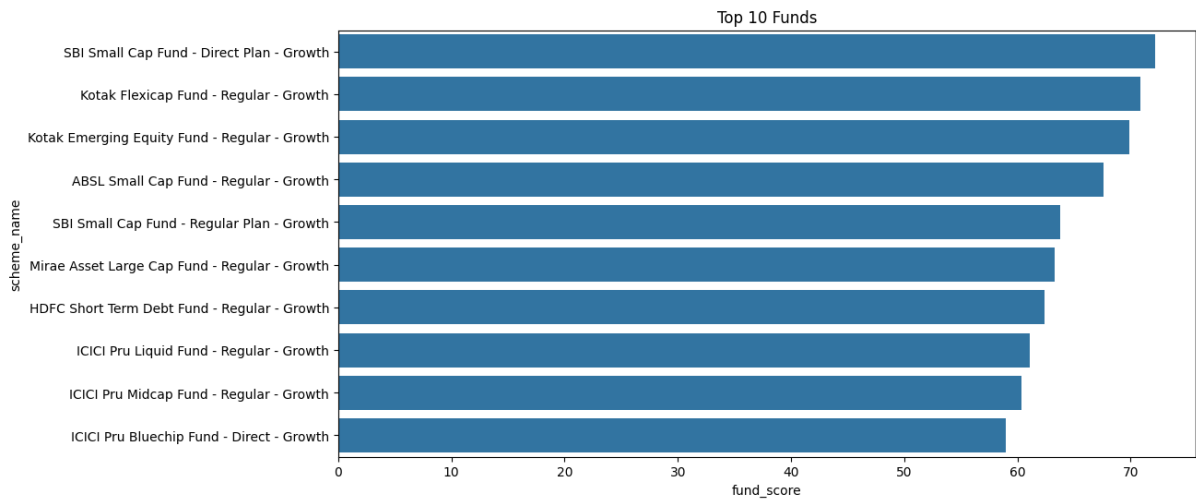
The following metrics were calculated:

- CAGR (Compound Annual Growth Rate)
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta

- Maximum Drawdown

These metrics helped identify top-performing funds and compare performance across schemes.

Key findings showed that risk-adjusted metrics provided more meaningful rankings than absolute returns alone.



Top 10 Fund Score Chart

Investor Analytics

Investor analytics focused on understanding demographic behavior and transaction patterns.

Analysis included:

- Age Group Distribution
- Gender Distribution
- City Tier Analysis
- State-wise Investment Analysis
- SIP versus Lumpsum Preferences

Key observations indicated that SIP transactions represented a significant share of investor activity and that investment behavior varied across demographic groups.

Benchmark Analysis

Mutual fund performance was compared against benchmark indices.

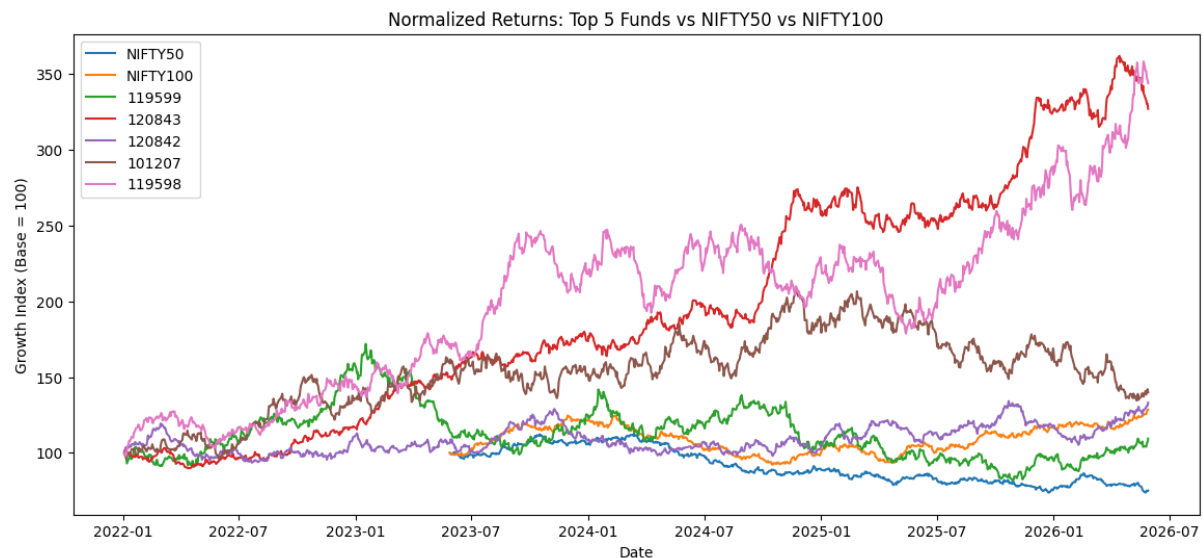
Benchmarks used:

- NIFTY 50
- NIFTY 100
- NIFTY Midcap 150

Benchmark analysis included:

- Relative Performance Comparison
- Tracking Error Analysis
- Normalized Return Comparison

The analysis highlighted differences in benchmark tracking efficiency across funds.



Benchmark Comparison Chart

Advanced Analytics

Advanced analytical techniques were implemented to evaluate portfolio risk and investor behavior.

Value at Risk (VaR)

Historical VaR at the 95% confidence level was calculated to estimate potential losses.

Conditional Value at Risk (CVaR)

CVaR was calculated to estimate expected losses beyond the VaR threshold.

Rolling Sharpe Ratio

A rolling 90-day Sharpe Ratio was used to measure risk-adjusted performance consistency over time.

Volatility Analysis

Volatility analysis identified funds with higher risk exposure.

Investor Cohort Analysis

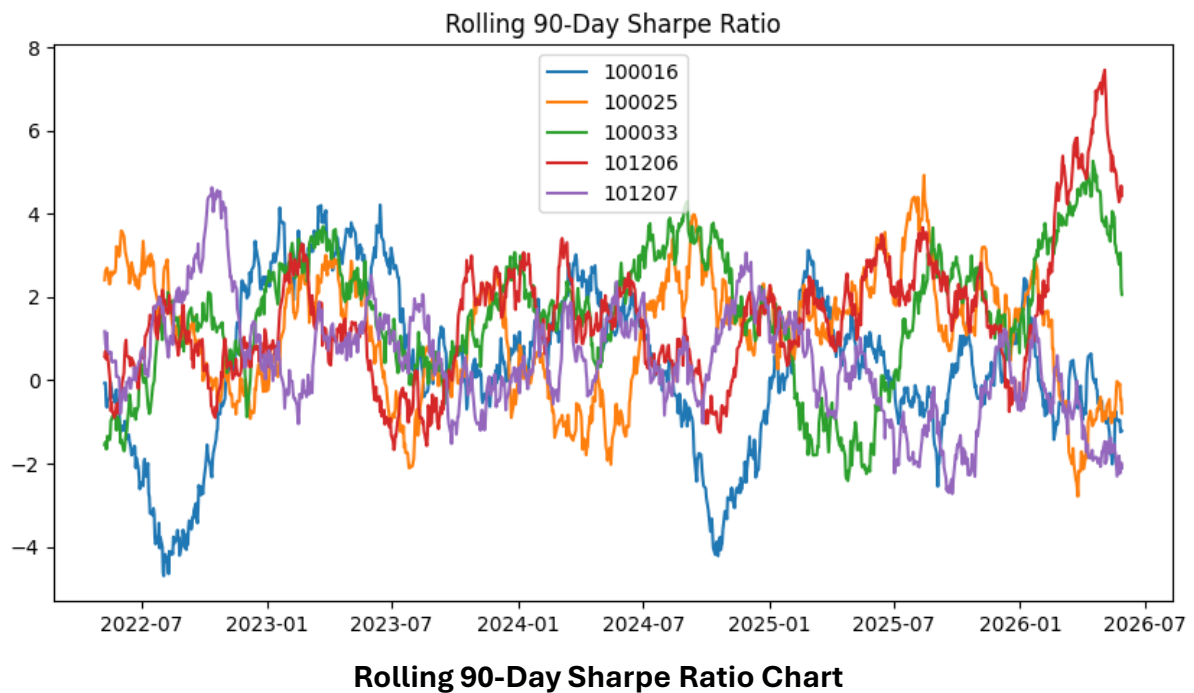
Investors were grouped based on their first transaction year to study investment patterns across cohorts.

SIP Continuity Analysis

Investor transaction gaps were analyzed to identify at-risk investors with inconsistent SIP contributions.

Fund Recommendation System

A rule-based recommendation model was developed using risk categories and Sharpe Ratios.



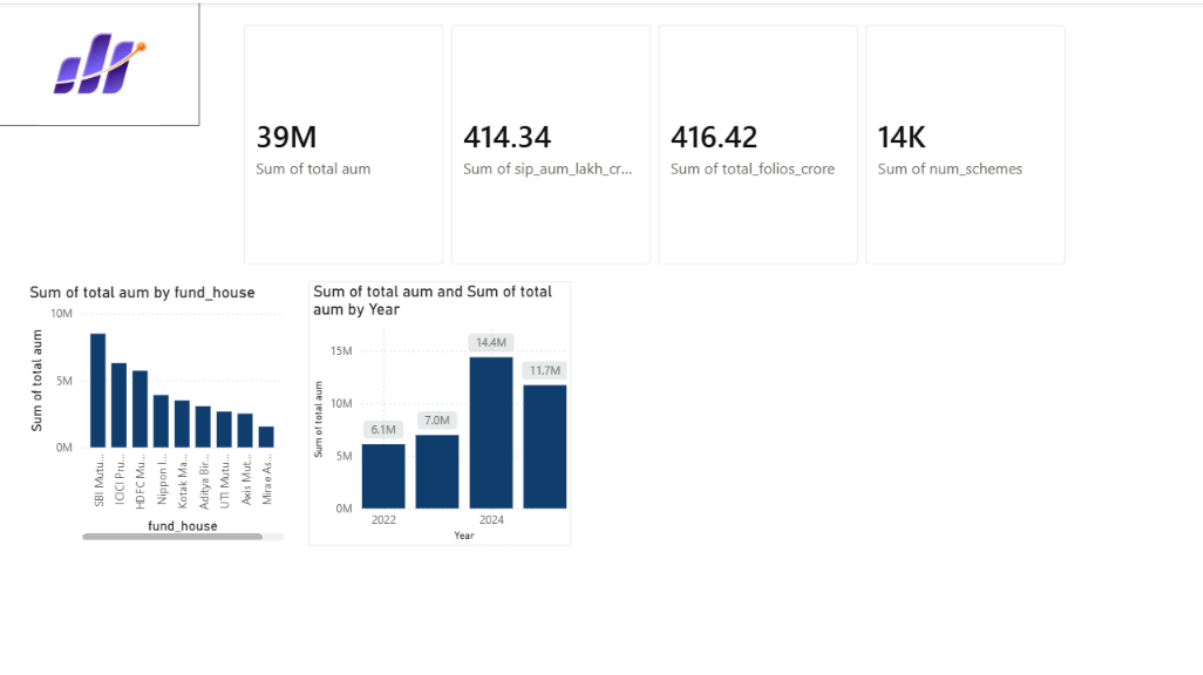
Dashboard Overview

An interactive Power BI dashboard was developed to visualize key metrics and insights.

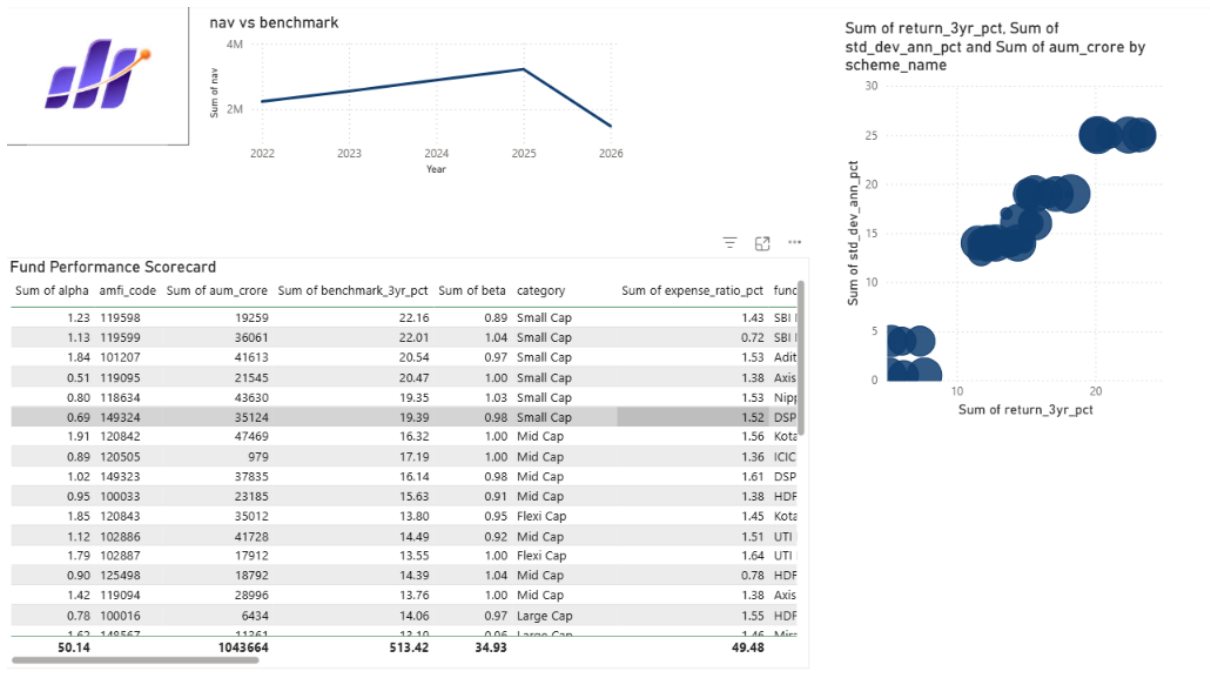
Dashboard pages include:

- 1. Industry Overview
- 2. Fund Performance
- 3. Investor Insights
- 4. Risk Analytics

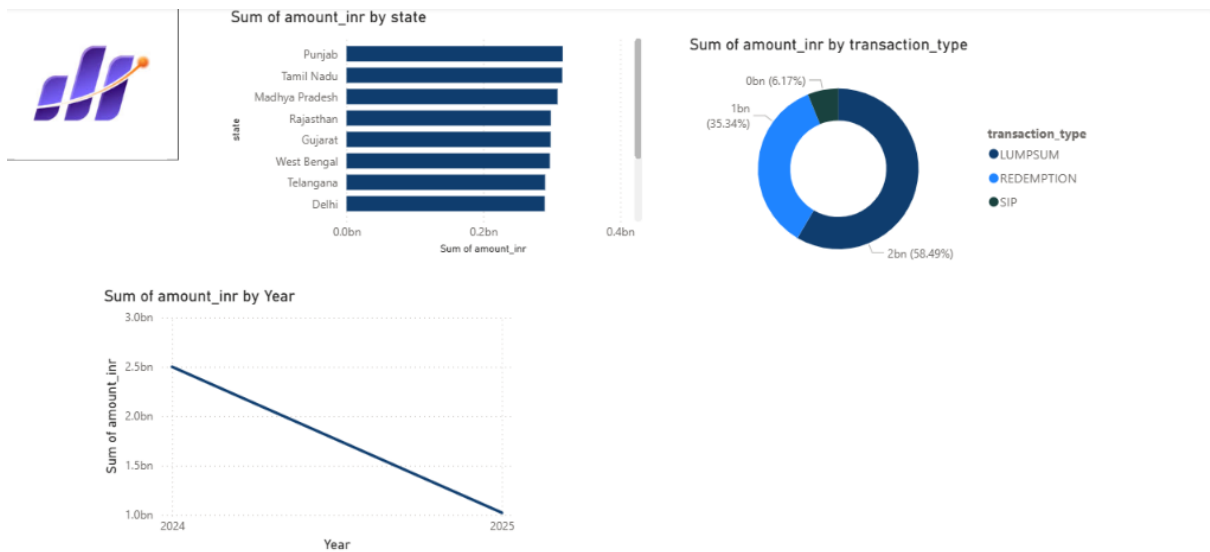
The dashboard provides users with an intuitive interface for exploring trends, performance indicators, and risk measures.



Industry Overview



Fund Performance



Investor Insights



Risk Analytics

Key Findings

Major findings from the project include:

- Top-performing funds achieved superior Sharpe Ratios.
- SIP transactions dominated investor activity.
- Benchmark tracking efficiency varied across funds.
- High volatility did not always lead to higher returns.
- Risk-adjusted metrics provided better performance evaluation.
- Investor behavior differed across age groups and geographic regions.

Recommendations

Based on the analysis, the following recommendations are proposed:

- Investors should prioritize risk-adjusted performance metrics.
- SIP continuity should be monitored to improve investor retention.
- Portfolio diversification should be encouraged to reduce concentration risk.
- Benchmark tracking should be reviewed regularly.

- Advanced risk metrics should be incorporated into investment decision-making processes.
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Limitations

The project has the following limitations:

- Historical data availability was limited.
 - Benchmark coverage may not represent all market conditions.
 - Certain assumptions were required during data cleaning and preprocessing.
 - Results are dependent on the quality of source data.
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Conclusion

The Mutual Fund Analytics Capstone Project successfully integrated data engineering, financial analytics, and business intelligence techniques into a unified solution.

The project demonstrated practical applications of Python, SQLite, and Power BI for analyzing mutual fund performance, investor behavior, benchmark tracking, and portfolio risk. The resulting insights can support informed investment decisions and improve understanding of mutual fund industry trends.

References

- Mutual Fund Industry Data
- Benchmark Index Data
- Python Documentation
- Pandas Documentation
- Power BI Documentation
- SQLite Documentation